

Suman Janarthanan

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EDUCATION

IIT GUWAHATI

DATA SCIENCE (CREDIT PROGRAM)
Expected Mar 2026

I.K.G. PTU, PUNJAB

B. TECH IN COMPUTER SCIENCE
Jun 2025 | CGPA: 7.71/10

GOVT MODEL HR SEC

Grad. April 2020 | JEE: 80%ile

SKILLS

ARTIFICIAL INTELLIGENCE

New Advanced:

GPT-4o • Gemini Pro 1.5 • Llama 3 • RAG
• LangChain • Pinecone • LangGraph •
Agentic AI • Vector DBs • WhisperAI •
Wav2Lip

Core Libraries:

Python • Transformers • Keras •
TensorFlow • PyTorch • Scikit-learn •
OpenCV • MTCNN • YOLOv8 • Face
Fusion • CNNs • NLP • Computer Vision

GenAI Tools:

Lobe.ai • Synthesia • DALL-E •
MidJourney • Fotor

TECHNICAL STACK

Backend DevOps:

Django REST • FastAPI • Docker • Nginx
• Gunicorn • Redis • Celery •
PostgreSQL • MongoDB • CI/CD •
Prometheus • Grafana

Infrastructure Other:

AWS (EC2, S3) • VPS • WebRTC •
WebSockets • IoT • Arduino • C/C++ •
FFmpeg • Git • Linux/Ubuntu

EXPERIENCE

DMX TECH SERVICES PVT LTD | AI ENGINEER & TECHNICAL TEAM LEAD

Feb 2025 – Present | Bangalore

- Leading 10+ cross-functional teams (10-25 engineers) to deliver production AI systems serving **15,000+ active users** with a **99.7% uptime SLA**.
- Architected a hybrid LLM strategy (GPT-4o + Llama 3) reducing operational costs by **85%** while maintaining a **0.847 BLEU score** for translation.
- Designed and enforced CI/CD pipelines and LLD standards, resulting in a **65% reduction** in production bugs and **40% increase** in team productivity.

SIRUSAI PRIVATE LIMITED | AI DEVELOPER

Aug 2023 – Oct 2024 | Dharmapuri

- Engineered a surveillance facial recognition system achieving **95% accuracy** using WebRTC for real-time P2P CCTV camera streaming.
- Improved feature extraction performance by **30%** through data augmentation and SVM training on a 1000+ image-per-person dataset.
- Deployed containerized models via Docker on VPS, handling real-time inference with optimized latency.

KEY PRODUCTION PROJECTS

PROJECT 1: CLOUD CODING COMPILER PLATFORM | DOCKER, RESTAPI, WEBSOCKETS

- Re-engineered infrastructure to scale from 100 to **5,000+ concurrent users** (**50x capacity increase**) while reducing costs by **58.5%**.
- Achieved a **95.7% faster** response time (**180ms p95 latency**) using Gunicorn optimization and Redis connection pooling.
- Implemented secure, sandboxed code execution for 20+ languages using ephemeral Docker containers with strict resource limits.

PROJECT 2: AI-POWERED INTERVIEW PLATFORM | LANGGRAPH, RAG, PINECONE

- Built an automated system handling **500+ interviews/day**, reducing time-to-hire by **75%** (from 4 weeks to 5 days).
- Developed an Agentic AI learning path using **LangGraph**, achieving **88% candidate engagement** compared to the 12% industry average.
- Reduced cost-per-interview from **\$80 to \$1.20 (98.5% savings)** using RAG-based resume analysis and dynamic question generation.

PROJECT 3: AI VIDEO TRANSLATION PIPELINE | WHISPERAI, WAV2LIP, NLLB-200

- Created an end-to-end pipeline that reduced processing time for a 1-hour video from **3 weeks to 4 hours (99% faster)**.
- Lowered translation costs by **98.3%** (\$5,600 to \$96 per hour) using a two-stage hybrid NMT architecture.

PHILOSOPHY

PRODUCTION-FIRST

Research is exciting, but real users need reliability. I build for 99.7% uptime.

SCALE-READY

If it works for 100, it must work for 100,000. I use horizontal autoscaling for all ML workloads.

COST-INNOVATION

Smart architecture beats expensive models. I prioritize hybrid LLM strategies to save 85% on API costs.

AWARDS

Nov 24 SIH (ISRO PS)
Sep 23 1st State Ideathon
May 23 2nd Udhyam Govt
Apr 23 1st Best Innovation
Mar 23 1st Prog2Entrepreneur

PUBLICATIONS

Neuralink: Advancing BMI for Enhanced Human-Machine Interaction | [Link](#)

PROJECT 4: SMART EXAM EVALUATION SYSTEM | COMPUTER VISION, OPENCV, GPT-4o

- Automated grading for **160+ centers** across India, processing **600,000+ answer sheets** in 8 months with **98.7% accuracy**.
- Reduced evaluation cycles from **15 days to 18 hours (90% reduction)** while saving 93.8% in evaluator fees.
- Engineered a hybrid OMR/OCR pipeline with a human-in-the-loop system for only 3% of edge cases.

PROJECT 5: INTELLIGENT DOCUMENT ANALYSIS | PADDLEOCR, CHROMADB, RAG

- Increased processing throughput **140x** (from 500 docs/week to 10,000 docs/day) with **93% faster** verification.
- Improved fraud detection rates to **96.8%** using Siamese neural networks for signature verification and pixel-level tampering analysis.
- Digitized documents for 8 universities, achieving **98.75% cost savings** per document verified.

PROJECT 6: AI COPYRIGHT DETECTION ENGINE | FFMPEG, SBERT, LLAMA 3

- Built a context-aware detection engine that reduced false positives by **24%** compared to industry-standard tools.
- Developed an LLM ensemble (Llama 3 + Mixtral) to reason through "Fair Use" cases, achieving **88.3% agreement** with legal reviewers[cite: 232, 237].
- Processed 1,000 videos in **45 minutes (96% faster)** using parallelized workers and acoustic hashing.

PROJECT 7: WORD-TO-DATABASE EXTRACTION | SENTENCE-BERT, SIMHASH, LLAMA 3

- Digitized **500,000+ questions** from legacy Word documents into a searchable PostgreSQL/Elasticsearch database.
- Achieved **99.1% precision** in semantic deduplication using SimHash and Sentence-BERT fusion.
- Increased throughput by **50x**, completing a 12-month manual project in just **2.5 months**.

HISTORICAL PROJECTS

HYBRID AUTOMATED GRASS CUTTER | VOICE RECOGNITION, IoT

- Integrated VOSK for offline voice recognition, reducing manual intervention by **80%**.

FACE FUSION (LOW LIGHT) | CNN, HoG

- Achieved **90%+ recognition accuracy** at distances up to 15 meters in low-light environments.